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MEASUREMENT AND UNCERTAINTY OF HEAT FLUX TO A RAIL-CASK SIZE PIPE CALORIMETER IN A POOL FIRE

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ABSTRACT

The goal of this work was to measure the temporally varying heat flux and surface temperature of a pipe calorimeter in a pool fire, and assess its uncertainty. Three large-scale fire tests were conducted at the Sandia National Laboratories outdoor fire test facility. In each test a cylindrical calorimeter was suspended above a water pool with JP8 fuel floating on top. The calorimeter was a 2.4 m diameter, 4.6 m long, and 2.5 cm wall thickness pipe with end-caps suspended 1 m above the 7.2 m diameter pool. 58 thermocouples were attached to the calorimeter interior surface and backed with 8 cm of insulation.

The Sandia One-Dimensional Direct and Inverse Thermal (SODDIT) code was used to determine the calorimeter external surface heat flux and temperature from the measured interior surface temperature versus time.

To determine the uncertainty of the SODDIT results, a simulation of the calorimeter in a fire similar to the experiments was performed using the Container Analysis Fire Environment (CAFE) computer code. In this code, a Computational Fluid Dynamics (CFD) fire model applies a temporally and spatially varying heat flux to the exterior surface of a Finite Element (FE) calorimeter model. Flux is similar but not identical to the flux in the experiment. The FE model calculates the internal calorimeter surface temperature, which is used by SODDIT to calculate heat flux which was compared to the applied values.

The absorbed heat flux and surface temperature at one calorimeter location was calculated by SODDIT and then compared to the CAFE applied heat flux and surface temperature. From this comparison a base case uncertainty due to inherent inverse calculation errors and frequency smoothing methods is presented.

Uncertainties in temperature measurements, calorimeter material properties and wall thickness were applied to the SODDIT calculation and iterated using the Monte Carlo method to determine the overall heat flux and surface temperature uncertainty. The total absorbed heat flux uncertainty at the one studied location is ±4.8 kW/m^2 at 95% confidence. The outer surface temperature uncertainty for all data at the one studied location is ±6.6°C at 95% confidence. For all 58 measurement locations, the overall combined total absorbed heat flux uncertainty is ±13.8 kW/m^2 at 95% confidence, surface temperature uncertainty is ±7.6°C. These uncertainties are valid only when the calorimeter temperature is not within the Curie temperature range of 999 to 1037K.

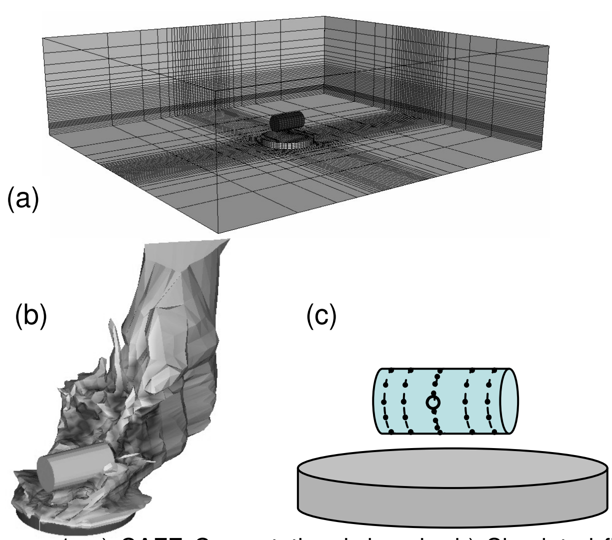

Figure 1. a) CAFE Computational domain, b) Simulated fire, c) Thermocouple locations

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INTRODUCTION

This work describes the method used to calculate absorbed heat flux from a pool fire to a rail-cask sized pipe calorimeter and especially its uncertainty.

Three large-scale fire tests were conducted at the Sandia National Laboratories outdoor fire test facility [1]. In each test a cylindrical calorimeter was suspended above a water pool. The calorimeter was a 2.4 m diameter, 4.6 m long, and 2.5 cm wall thickness pipe with end-caps suspended 1 m above the 7.2 m diameter pool with 7.6 m3 (2000 gallons) of JP8 fuel floating on top.

Type-K thermocouples were strap welded to the calorimeter interior surface and backed with 8 cm of insulation. 48 of these were located on equally spaced rings on the pipe calorimeter body, and five were on each end cap. The temperatures of all thermocouples were acquired once per second during and after the fire. The wind speed, direction and temperature were measured by anemometers on four poles that surrounded the facility, at three elevations each. The flame shape and motion were recorded by video from four different angles. The average wind speeds for the three tests were 0.8 m/s, 1.1 m/s, and 2.6 m/s.

The Sandia One-Dimensional Direct and Inverse Thermal (SODDIT) code [2] was used to determine the calorimeter external surface heat flux and temperature from the measured interior surface temperature versus time. The SODDIT inverse conduction code calculates absorbed heat flux when given interior surface temperature versus time data as well as other inputs such as material thermal properties, domain dimensions, and other parameters which are used to stabilize the inverse conduction algorithm.

The way in which uncertainties in inputs contribute to uncertainty in the calculated absorbed heat flux is complicated and has been studied previously [3,4]. Concurrent work has yielded the Container Analysis Fire Environment (CAFE) computer code [5]. In this work the CAFE tool is used to help quantify SODDIT heat flux uncertainty.

The CAFE code contains CFD, transport, and radiation models for the fire region as well as a One-Dimensional conduction model to simulate the response of the calorimeter itself. It has been shown that the CAFE predicted temperature rise of an engulfed calorimeter was similar to actual fire experiments [1].

The CAFE model calculates both the applied heat flux and the interior temperature, SODDIT was used to calculate heat flux from the CAFE supplied interior temperature and a comparison was made to the CAFE applied heat flux. In this way, in addition to input parameter variation, an estimate of the absorbed heat flux uncertainty was obtained.

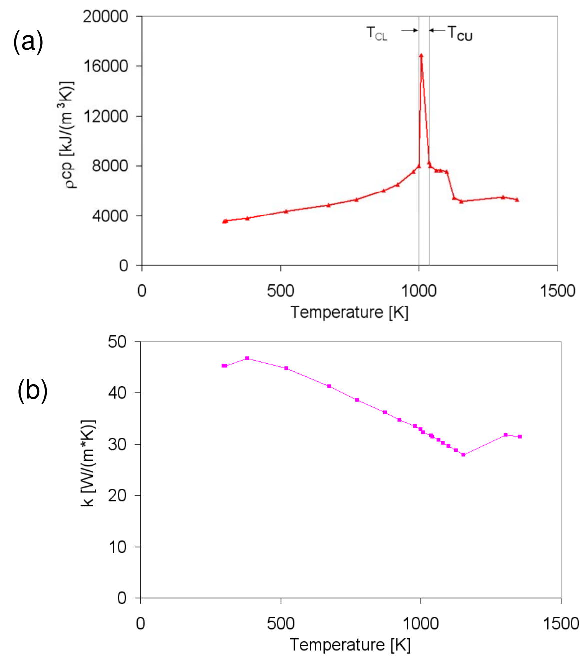

Figure 2. Carbon steel calorimeter properties, a) Density × specific heat versus temperature, b) Thermal conductivity versus temperature

NOMENCLATURE

d	calorimeter wall thickness
$d_M(i)$	simulated wall thickness with errors added
i	Monte Carlo iteration number, i = 1 to n_M
j	temperature data sample number, j = 1 to n_t
k	carbon steel thermal conductivity
$k_M(T,i)$	simulated thermal conductivity with errors added
n_M	Number of Monte Carlo iterations
n_t	number of times temperature is sampled
$q''_{O,C}$	CAFE outer surface heat flux
$q''_{O,C,20s}$	CAFE outer surface heat flux, 20 second window average
$q''_{O,S}$	SODDIT outer surface heat flux
t_{CI}	time when Curie temperature range is entered
t_{CF}	time when Curie temperature range is left
t_j	measurement times = $\Delta t \times (j-1)$
B_{Ax}	bias error amplitude, as a fraction of the true value (different for each variable)
$E_{q''}$	heat flux error
E_T	surface temperature error
NFT	number of future times of SODDIT calculation
R(i,j) and R(i)	random number series with standard deviation equal to 1 and mean equal to 0.
R_A	random error amplitude
$T_{I,C}$	CAFE predicted inner surface temperature
T_{CU}	upper Curie temperature range
T_{CL}	lower Curie temperature range

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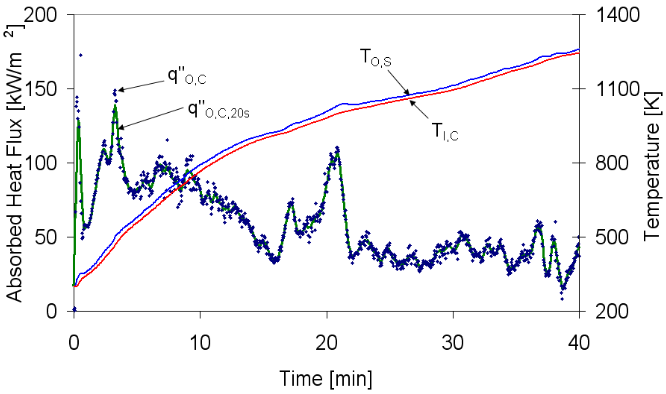

Figure 3. CAFE calculated heat flux (both 20 second window averaged and not window averaged) and Interior temperature versus time

$T_M(i,t_j)$	simulated measured temperature with errors added
$T_{O,S}$	SODDIT predicted outer surface temperature
$T_{O,C}$	CAFE outer surface temperature
$T_T(t_j)$	true temperature of inner surface at time t_j
Δt	sampling time interval
ρc_p	steel density × specific heat (volumetric specific heat)
$\rho c_{pM}(T,i)$	simulated measured density × specific heat with errors added
σ	standard deviation

BASE CASE UNCERTAINTY

For simplicity, this work focuses on one location on the calorimeter, the North side location of the middle section. This location was engulfed for most of the fire experiment since the wind blew primarily from East to West during the test.

Figure 1a shows the CAFE computational domain with the calorimeter and fire pool shown in the center. Figure 1b shows a snapshot of the CAFE simulated fire, which resulted in calorimeter temperature rise similar to the first experiment when given identical wind boundary conditions and fuel mass injection rate. Figure 1c shows the test calorimeter and the locations where thermocouples were attached to the interior surface. The primary location discussed in this work is highlighted.

The CAFE and the SODDIT codes require material properties to be supplied. Figure 2 shows the material properties which were used for the calorimeter carbon steel. These properties were measured from a steel sample of the same type and vendor as the calorimeter steel. Of particular interest in Figure 2a is the peak in ρc_p between 999 and 1037K. This peak is due to the Curie phase change which occurs in carbon steels. The phase change actually occurs at an exact temperature which is dependent on the alloy, not through a range of temperatures. The 38K wide peak is an artifact of the Differential Scanning Calorimetry (DSC) technique which was used to measure ρc_p. Because of this phase change and the trouble it causes inverse conduction codes [6], many fire test calorimeters are made from stainless steel instead of carbon steel. Carbon steel was used for these tests due to the large size and budget constraints. Carbon steel thermal conductivity, shown in Figure 2b, varies more smoothly with temperature and does not cause a problem.

Figure 3 shows the absorbed heat flux calculated by the CAFE simulation versus time. Symbols represent $q''_{O,C}$, the CAFE outer surface absorbed heat flux. The bold line represents $q''_{O,C,20s}$, a 20 second window average of the CAFE absorbed heat flux. $q''_{O,C,20s}$ is plotted at 5 second time intervals, while $q''_{O,C}$ was generated at various time intervals averaging 1.5 seconds. A window average was used to smooth the high-frequency aspect of the heat flux so it could be compared directly against SODDIT calculated heat flux. Inverse conduction codes use future time data which have a smoothing effect on the predicted heat flux, much like a window average [3]. The number of SODDIT future times was chosen to be equal to 20 seconds of time (NFT = 4 since Δt = 5 sec), and the resulting heat flux high-frequency content is similar to the CAFE window averaged data. The 20 second window size is arbitrary, however the inverse code will become unstable if the future time is too small, and 20 seconds shows enough high-frequency content for the type of analysis the heat flux data is intended for.

Figure 3 also shows the CAFE inner surface temperature $T_{I,C}$ as well as the SODDIT calculated outer surface temperature $T_{O,S}$ plotted versus time. No time averaging of the temperature data was necessary since temperature is basically an integration of absorbed heat flux over time and does not change abruptly.

Figure 4 shows $q''_{O,C,20s}$ again and adds the SODDIT calculated exterior heat flux $q''_{O,S}$. Outer temperature $T_{O,S}$ is also shown. $q''_{O,S}$ tracks $q''_{O,C,20s}$ quite closely until t_{CI} = 19 minutes, at which time the calorimeter starts to enter the Curie affected temperature range between T_{CL} = 999K and T_{CU} = 1037K. From 19 minutes to t_{CF} = 24 minutes, $q''_{O,S}$ shows some major deviations from $q''_{O,C,20s}$. The instability occurs despite the fact that the CAFE and SODDIT numerical

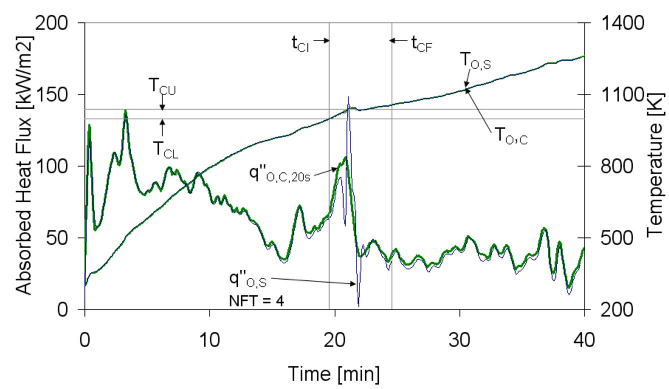

Figure 4. SODDIT heat flux and exterior temperature compared with CAFE heat flux and temperature versus time

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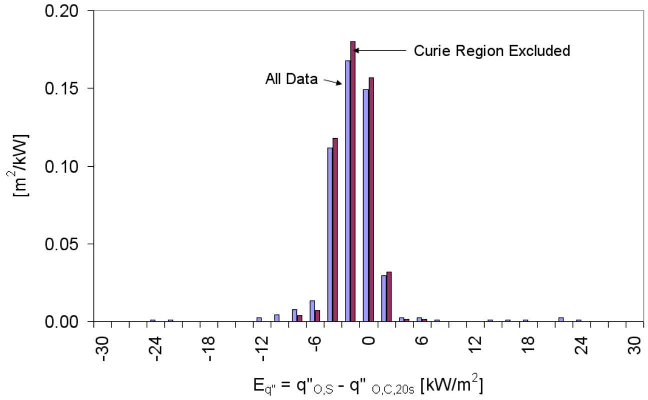

Figure 5. “Base Case” Heat flux error probability density function

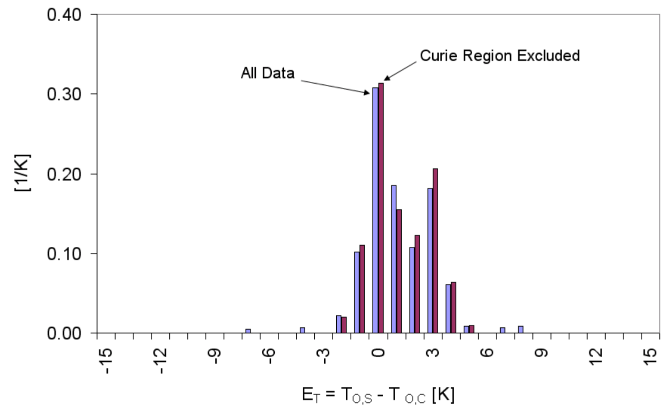

Figure 6. “Base Case” surface temperature error probability density function

models are using identical material properties. After 24 minutes, $q''_{O,S}$ follows $q''_{O,C,20s}$ reasonably well again.

The differences in the heat fluxes in Fig 4 represent the error of the SODDIT calculation due to inverse conduction numerical issues [3], and high frequency smoothing techniques. The error $E_{q''} = q''_{O,S} - q''_{O,C,20s}$ is shown in the form of a probability density function in Figure 5. This will be called the Base Case error, as it includes no temperature or material property uncertainties. The lighter bars represent errors from all data. The average error is -1.7 kW/m^2 and the standard deviation (2σ) is 13.2 kW/m^2. The darker bars exclude data from when the calorimeter temperature was within the Curie range. For the Curie excluded data the error averaged -1.6 kW/m^2 with standard deviation of 3.7 kW/m^2. These values are tabulated in Table 2.

Also shown in Figure 4 is the CAFE outer temperature $T_{O,C}$ and SODDIT calculated outer temperature $T_{O,S}$. These appear as one line since the SODDIT outer temperature calculation is quite good. The error $E_T = T_{O,S} - T_{O,C}$ is presented in the form of a probability density function in figure 6. Error standard deviation (2σ) was less than 4°C for all cases.

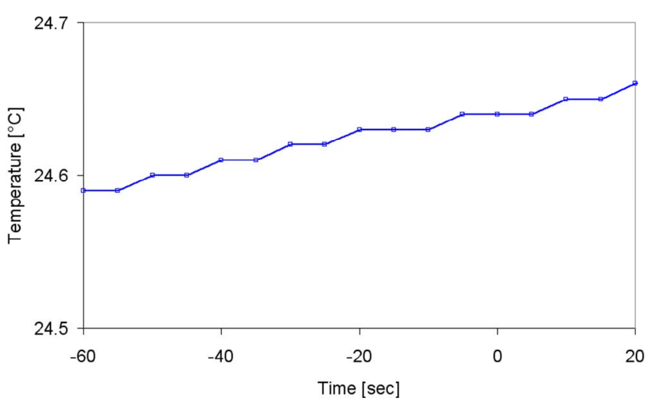

Figure 7. Thermocouple leader data temperature versus time

TEMPERATURE CONTIRBUTION

Since the calculation of heat flux from temperature data is related to the instantaneous slope of the temperature versus time curve, noisy temperature data can greatly contribute to heat flux error. The base case simulation used CAFE simulated temperature data with no real thermocouple errors. To obtain a reasonable estimate of the actual experiment thermocouple random noise errors, fire test data was examined. Figure 7 shows temperature versus time data from the real fire test, before the fire was lit. The temperature axis of the graph covers 0.2°C. The resolution of the data acquisition system was 0.01°C and no random noise is visible. This implies the random thermocouple noise is less than 0.01°C. This agrees with previous observations [6]. ±0.02°C was assumed at 95% confidence.

The ANSI standard MC96.1 for standard type K thermocouples specifies a tolerance of 0.75% bias error, referenced at 0°C. This amount of error was used for the following study and 95% confidence was assumed.

MONTE CARLO SIMULATION

If the previously mentioned random and bias thermocouple errors are superimposed onto the $T_{I,C}$ data, which is then used as input to SODDIT, the new heat flux error with temperature errors included can be determined. A Monte Carlo simulation was used to determine the effect of random and bias errors on the heat flux calculation uncertainty. In this type of uncertainty analysis, measurement errors are simulated using a random number generator and the calculation is done many times with different random measurement errors which approximate the unknown actual measurement errors. The resulting heat flux errors of the many iterations are then statistically analyzed to determine their standard deviation and mean values.

The simulation generates a new random error for each temperature data point (at every time sample j), as well as a new bias error for each iteration i. The random and bias errors were generated using a random number generator, normally

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Table 1. Input parameter uncertainties, 95% confidence

	Random Error, R	Bias Error, B
T	0.02°C	0.75%
ρc_p	0	2.0%
k	0	4.0%
d	0	4.0%

distributed, with 95% confidence level (2 standard deviations) equal to the temperature uncertainties stated previously and summarized in Table 1. The exact way in which the errors were added to the temperature data is detailed here:

Simulated measured temperature with errors added

$$T_M(i,t_j) = T_T(t_j)[1 + B_{AT}R(i)] + R_{AT}R(i,j)$$

The result of the Monte Carlo simulation with temperature errors shows that heat flux error 2σ deviation grows only slightly from 3.7 kW/m^2 to 3.9 kW/m^2 (excluding Curie range, all results are summarized in Table 2).

To determine the number of iterations required to come to convergence on a unique solution, a thousand SODDIT calculations were run using an automated computer program. The result of the convergence test for standard deviation is shown in Fig 7. The test determined that about 200 iterations are enough to provide a reasonable level of standard deviation precision and computer run time. 200 iterations were used for subsequent Monte Carlo simulations.

Table 2. One Location Absorbed heat flux and exterior temperature uncertainty

	Heat Flux [kW/m^2]		Exterior Temperature [°C]	
All Data	Average	2σ (95%)	Average	2σ (95%)
Base Case	-1.7	±13.2	1.1	±3.7
incl. Temp Errors	-1.7	±15.1	1.0	±6.8
incl. all Errors	-1.8	±15.7	0.9	±7.1
Curie Data Excluded	Average	2σ (95%)	Average	2σ (95%)
Base Case	-1.6	±3.7	1.2	±3.2
incl. Temp Errors	-1.7	±3.9	1.0	±6.2
incl. all Errors	-1.7	±4.8	0.9	±6.6

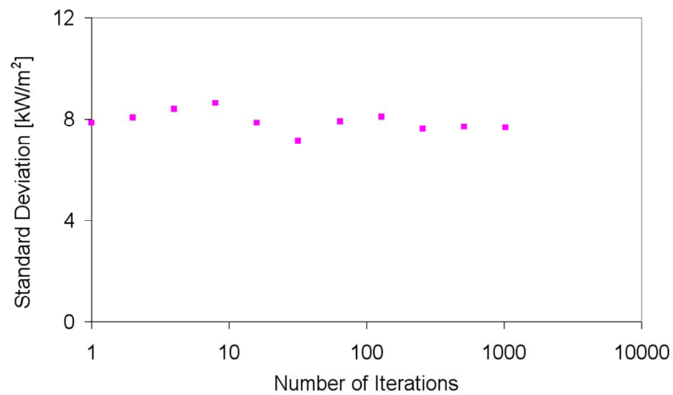

Figure 8. Monte Carlo heat flux standard deviation versus number of iterations

OTHER CONTRIBUTIONS

A previous study [4] found that the major contributors of uncertainty to inverse conduction calorimeter heat flux calculations are: temperature errors, calorimeter material density and specific heat errors, calorimeter wall thickness dimension errors. Insulation properties were not significant contributors. In this study interior insulation was modeled but not considered for the uncertainty analysis.

The Monte Carlo simulation used to determine SODDIT uncertainty contribution from temperature errors was revised to include errors from calorimeter material density and specific heat, thermal conductivity, and wall thickness. The uncertainties in these quantities at 95% confidence level are shown in Table 1. These uncertainty quantities were decided based upon engineering judgment or in the case of thermal properties, as reported by the test technician. Each of these uncertainties were treated as bias errors, where a new bias error for that parameter was generated for each iteration. In the case of ρc_p and k which are functions of temperature, the entire curve was shifted up or down by the error amount.

The exact way in which errors were added to material properties is detailed here:

Simulated measured density × specific heat, thermal conductivity, and calorimeter wall thickness with errors added

$$\rho c_{pM}(T,i) = \rho c_p(T)[1+ B_{A\rho cp}R(i)]$$ Density × Specific Heat

$$k_M(T,i) = k(T)[1+ B_{Ak}R(i)]$$ Thermal Conductivity

$$d_M(i) = d\,[1+ B_{Ad}R(i)]$$ Calorimeter Wall Thickness

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The results of the Monte Carlo simulation including all uncertainties are tabulated in Table 2. The 2σ heat flux uncertainty for the time when the calorimeter temperature was in the Curie range is excluded, is ±4.8 kW/m^2. It is interesting to note that the heat flux uncertainty showed only a small increase when temperature errors were applied. This is probably due to the larger uncertainty due to inverse conduction numerical issues, and the way uncorrelated uncertainties combine in a Root Sum Squares manner which accentuates the dominant uncertainty source and minimizes the effect of smaller sources [7].

SPATIAL VARIATION

Up to this point only one location on the calorimeter has been discussed. The same procedure can be used to determine uncertainty for each thermocouple location on the calorimeter. Figure 9 shows a contour plot of 2σ heat flux uncertainty, 95% confidence. The previously mentioned studied location is circled. The uncertainty varies with location on the calorimeter because some locations received more steady heat flux than others. For example the end caps, especially the West end, showed much higher uncertainty than other locations, probably because these areas were only intermittently engulfed, and were often exposed to cool surroundings between rising hot plumes (the end caps are not shown in figure 9). The places with least uncertainty are toward the calorimeter center and at some angle (not horizontal or vertical surfaces). The overall combined heat flux uncertainty for all locations (excluding the Curie temperature range) is ±13.8 kW/m^2 (95% confidence). If the end caps are excluded from the calculation the uncertainty drops to ±6.9 kW/m^2. Table 3 summarizes these results and also shows the surface temperature combined uncertainties.

CONCLUSION

It was found that for this case, the calorimeter absorbed heat flux uncertainty contribution caused by SODDIT numerical issues and smoothing is greater than the contribution due to uncertainties in other variables such as temperature, material properties, and wall thickness. The greatest contributor of heat flux uncertainty is due to the Curie phase change of carbon steel. Fortunately, this effect only lasts a short period of time for most locations and occurs after much of the period of high heat flux is over. The total absorbed heat flux uncertainty for all locations is ±13.8 kW/m^2 at 95% confidence (time when the calorimeter temperature was in the Curie range is excluded). Outer surface temperature uncertainty is ±7.6°C.

Table 3. All locations combined uncertainty

	Curie Data Excluded			
	Heat Flux [kW/m^2]		Exterior Temperature [°C]	
incl all errors	Average	2σ (95%)	Average	2σ (95%)
incl. End Caps	-1.2	±13.8	0.9	±7.6
excl. End Caps	-1.1	±6.9	1.0	±5.6

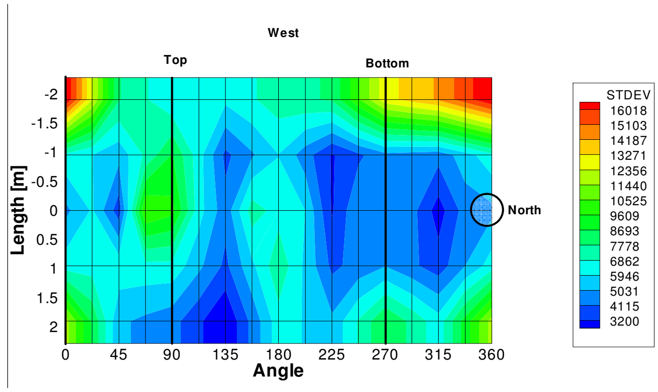

Figure 9. 2σ uncertainty contour plot of the calorimeter body. Units are W/m^2

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